VISHAL BAKSHI

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Education

George Washington University — **Aug 2023 – May 2025**
Masters of Science in Data Science — *Washington, DC*

Jawaharlal Nehru Technological University — **June 2018 – Aug 2022**
Bachelors of Technology in Electronics and Communication Engineering — *Hyderabad, India*

Experience

George Washington University — **May 2024 – May 2025**
Machine Learning Engineer — *Washington, DC*

- Developed and implemented an ELT pipeline to extract, preprocess, and analyze 2.01 TB of patient data (1,067 records) using Python, optimizing data workflows for predictive diabetes risk modeling in the AI-READI project.
- Integrated 5+ time-based features from auxiliary datasets to improve overall model performance, resulting in a 20% boost in recall.
- Collaborated with interdisciplinary teams and subject matter experts to integrate best practices, ensuring robust data-driven research and actionable clinical insights.

HBots Robotics Pvt Ltd — **Jun 2022 – Aug 2023**
Machine Learning Engineer — *Hyderabad, Telangana*

- Developed and deployed a real-time object detection model using MobileNet V1 in PyTorch, achieving a 35% boost in mAP by building a custom product dataset with varied angles and lighting, and applying targeted data augmentation for improved inventory tracking.
- Led a cross-functional ML project by integrating Intel RealSense stereo depth cameras for volume estimation, enabling accurate price extraction by complementing template-based object detection with real-world dimension analysis.
- Collaborated with team members using version control systems such as Git to organize modifications and assign tasks.

Projects

Deep Fake Video Classification | *Python, AWS EC2, Tensorflow, Computer Vision* — **October 2024**

- Developed a deep fake detection pipeline by extracting frames with OpenCV, addressing class imbalance through undersampling, and applying Error Level Analysis (ELA) and Discrete Fourier Transform (DFT) for feature extraction.
- Fused InceptionV3-extracted 1024D ELA features with a 512D azimuthal-averaged DFT array, normalizing inputs and leveraging an SVM with RBF kernel to enhance deepfake classification.
- Improved F1 score from 0.45 to 0.80 by implementing advanced preprocessing, feature fusion, and optimized classification techniques, accelerating computations on AWS EC2 GPU instances.

New York Airbnb Review Analysis | *AWS EC2, NLTK, Spacy, LLM's, Scikit-Learn, PyTorch* — **November 2024**

- Developed and fine-tuned a BERT-based sentiment analysis model with an F1 score of over 90%, analyzing 17,444 Airbnb reviews to classify sentiment with high accuracy and provide actionable insights into guest satisfaction.
- Engineered an advanced NLP pipeline utilizing T5 for summarization, generating concise summaries from large text datasets; streamlined review analysis for faster interpretation of key themes like host interaction, cleanliness, and location.

Bank Churn Analysis using AWS | *Python, AWS Sagemaker, S3, Quicksight* — **December 2024**

- Built a Bank Churn Analysis model using AWS SageMaker, achieving 98% accuracy through robust preprocessing, RFE feature selection, and RandomForest, optimizing customer retention predictions.
- Integrated AWS QuickSight for advanced visualization and streamlined the workflow using SageMaker for scalable training and S3 for efficient storage, ensuring seamless deployment and data-driven decision-making.

Technical Skills

Languages: Python, R, SQL, Shell scripting
Big Data Cloud: AWS, Spark, Hadoop
Databases: MySQL, PostgreSQL, MongoDB
Visualization Tools: Tableau, Seaborn, Plotly, Matplotlib, AWS Quicksight
Devops: Git, GitHub, CI/CD, Docker
Frameworks: Streamlit, SpaCy, NLTK, TensorFlow, PyTorch, ScikitLearn, OpenCV, Keras, LangChain, langGraph